SUBSYSTEM ONLY — DO NOT FABRICATE AS COMPLETE MP3 PLAYER.
Power supply, charger, headphone amplifier, USB and SD socket circuits are external interfaces here.
Pin-level connectivity is checked; electrical ratings and the full product remain under review.

U1
ESP32-WROVER
ESP_EN
I2C_SDA
I2C_SCL
STRAP_GPIO12
SD_CS
AMP_ENABLE
STRAP_GPIO15
BATT_SENSE
IO_INT
BTN_PWR
BTN_PLAY
UART_TX
UART_RX
ESP_BOOT
STRAP_GPIO2
TFT_BL
TFT_CS
SPI_SCK
SPI_MISO
TFT_DC
TFT_RST
SPI_MOSI
I2S_WS
I2S_BCK
I2S_DATA
R1
10k
ESP_EN
R2
10k
ESP_BOOT
R3
10k
STRAP_GPIO12
R4
10k
STRAP_GPIO2
R5
10k
STRAP_GPIO15
J2
GC9A01 MODULE INTERFACE
+3V3
GND
SPI_MOSI
SPI_SCK
TFT_CS
TFT_DC
TFT_RST
TFT_BL
R6
10k
TFT_CS
R7
10k
SD_CS
R8
10k
TFT_RST
R9
100k
AMP_ENABLE
R10
100k
TFT_BL
J1
REGULATED 3V3 / GND / BAT SENSE INPUT
VBAT_SENSE
C1
1uF / 10V
C2
100nF / 10V
C3
10uF / 10V
SW8
RESET
SW9
BOOT
J3
3V3 LOGIC UART / GND
R11
4.7k
I2C_SDA
R12
4.7k
I2C_SCL
R13
10k
IO_INT
R14
10k
BTN_PWR
R15
10k
BTN_PLAY
J4
I2S TO PENDING AUDIO SHEET
I2S_BCK
I2S_WS
I2S_DATA
AMP_ENABLE
U4
TCA9554APWR
R16
10k
BTN_NEXT
R17
10k
BTN_PREV
R18
10k
BTN_VOLUP
R19
10k
BTN_VOLDN
R20
10k
BTN_MODE
J5
CHARGER / CARD / HEADPHONE STATUS
CHG_STAT
SD_CD
HP_DET
R21
10k
CHG_STAT
R22
10k
SD_CD
R23
10k
HP_DET
C4
100nF / 10V
J6
3V3 SPI MICROSD INTERFACE
SPI_MISO
SPI_SCK
SD_CS
SW1
BTN_PWR
SW2
BTN_PLAY
SW3
BTN_NEXT
SW4
BTN_PREV
SW5
BTN_VOLUP
SW6
BTN_VOLDN
SW7
BTN_MODE
C5
100nF / 10V
C6
100nF / 10V
R24
100k 1%
VBAT_SENSE_IN
R25
100k 1%
R26
10k
#FLG1
PWR_FLAG
#FLG2
PWR_FLAG

External power, audio, USB and SD interfaces; full product pending datasheet verification

Sheet: /
File: digital_core.kicad_sch
Title: MP3 digital core — SUBSYSTEM ONLY
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